

Eric Chen

Berkeley, CA | [Portfolio](#) | [LinkedIn](#) | 248-854-9485 | beiyuanchen@gmail.com

EDUCATION

- **University of California, Berkeley, Astrophysics** *Aug 2026 - Expected May 2028*
- **University of California, San Diego, GPA: 3.71, Astrophysics** *Sep 2024 - Jun 2026*

EXPERIENCE

- Lawrence Berkeley Lab - Lazuli Space Observatory, Undergraduate Researcher** *Sep 2026 - Present*
- Modeling spacecraft vibration and how it affects telescope image quality and wavefront sensing.
 - Simulating reaction-wheel disturbances and analyzing their impact on optical measurements.

- UC Berkeley - UltraViolet EXplorer (UVEX), Undergraduate Researcher** *Sep 2026 - Present*
- Mapping UV transient duration-luminosity phase space from literature data to identify underexplored discovery regions for UVEX.

- Space Enterprise at Berkeley, Tracking & Controls Engineer** *Sep 2026 - Present*
- Improving rocket position tracking using GPS, motion-sensor, and altitude data.
 - Analyzing pointing error to help a ground radio antenna follow the rocket during flight.

- NASA Goddard - Roman Space Telescope, Integrated Modeling Intern** *Jun 2026 - Aug 2026*
- Implemented real-time log streaming for the Roman Space Telescope's Structural, Thermal, and Optical Performance (STOP) pipeline, enabling continuous remote monitoring of simulation progress.
 - Optimized MATLAB-based JittTool structural dynamics and jitter analysis workflows, reducing analysis runtime by 50%.

- Astrophysics Research - UCSD, Undergraduate Researcher** *Oct 2025 - Present*
- Developed neural-network models for 5-day geomagnetic activity forecasting, achieving ~85% recall with consistent performance across the forecast horizon.
 - Built data pipelines to process NASA datasets, including automated data collection and image processing.

- Astrophysics Research - Tsinghua University, Research Intern** *Jun 2025 - Sep 2025*
- Developed a U-Net-based machine-learning pipeline to identify H I shell candidates in THINGS radio astronomy data.
 - Achieved ~100% per-slice recall on held-out data for H I shell candidate screening.

- Foxit Software, LLM Integration Intern** *Jun 2025 - Sep 2025*
- Built an MCP bridge connecting Qwen to Foxit APIs for multi-step LLM tool execution.
 - Developed a local evaluation framework for testing tool selection and workflow orchestration.

LEADERSHIP

- LIVE AI Best Coast Hackathon (UCSD Location), Organizer** *Jul 2025 - Present*
- Recruited and managed cross-campus teams of speakers, panelists, judges, and student organizers.
 - Coordinated logistics for 100+ participants, including venue, catering, and UCSD approvals.

HONORS & AWARDS

- Co-author, 46th COSPAR Scientific Assembly, Florence, Italy

SKILLS

Programming: Python (advanced), MATLAB (advanced), C++ (intermediate), Java (intermediate), JavaScript

Scientific/ML: PyTorch, TensorFlow, NumPy, SciPy, Astropy, Git, Linux, LaTeX

Languages: English (Native), Mandarin Chinese (Native)